

Task 3: Web Application Penetration Testing

Internship Report

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Introduction

As part of the Cyber Security Internship Program, I performed a basic web application penetration testing exercise using Burp Suite Community Edition and OWASP Juice Shop. The objective of this task was to understand how web applications communicate over HTTP and to gain practical experience in intercepting, analyzing, and replaying web requests.

Tools Used

- Burp Suite Community Edition
- Docker Desktop
- OWASP Juice Shop
- Web Browser (Burp Browser/Chrome)

Environment Setup

Docker Desktop was installed and configured successfully. OWASP Juice Shop was deployed locally using Docker and made accessible through <http://localhost:3000>. Burp Suite was then launched to monitor and analyze the web traffic generated by the application.

Methodology

1. Launched OWASP Juice Shop using Docker.
2. Opened the application through a browser and verified accessibility.
3. Started Burp Suite and configured traffic interception.
4. Observed requests and responses in HTTP History.
5. Explored the application's structure through the Site Map feature.
6. Selected requests and examined their contents.
7. Sent requests to Repeater and replayed them to analyze server responses.

Observations

The application generated multiple HTTP requests and responses that were successfully captured by Burp Suite. The Site Map feature identified several application endpoints. Requests were replayed using Repeater, and valid responses were received from the server.

Results

The web application penetration testing activity was completed successfully. Traffic interception, endpoint discovery, request inspection, and response analysis were performed using industry-standard tools.

Conclusion

This task provided hands-on experience with web application penetration testing. By using Burp Suite and OWASP Juice Shop, I learned how to capture traffic, map application resources, inspect requests, and analyze server responses.

Attachments

Insert screenshots of Juice Shop Homepage, Site Map, HTTP History, Request Details, Repeater Request, Repeater Response, and Docker/Server Running Screen.